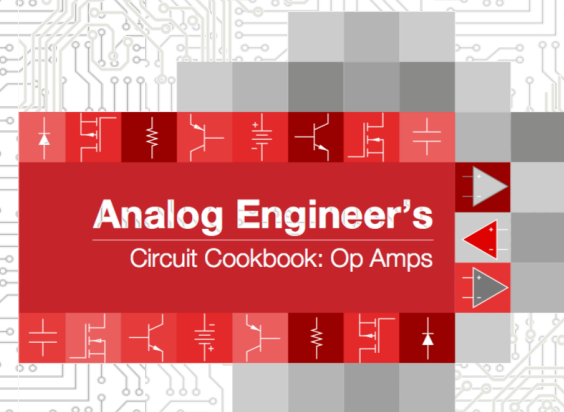


How to Design Temperature sensing with NTC thermistor circuit

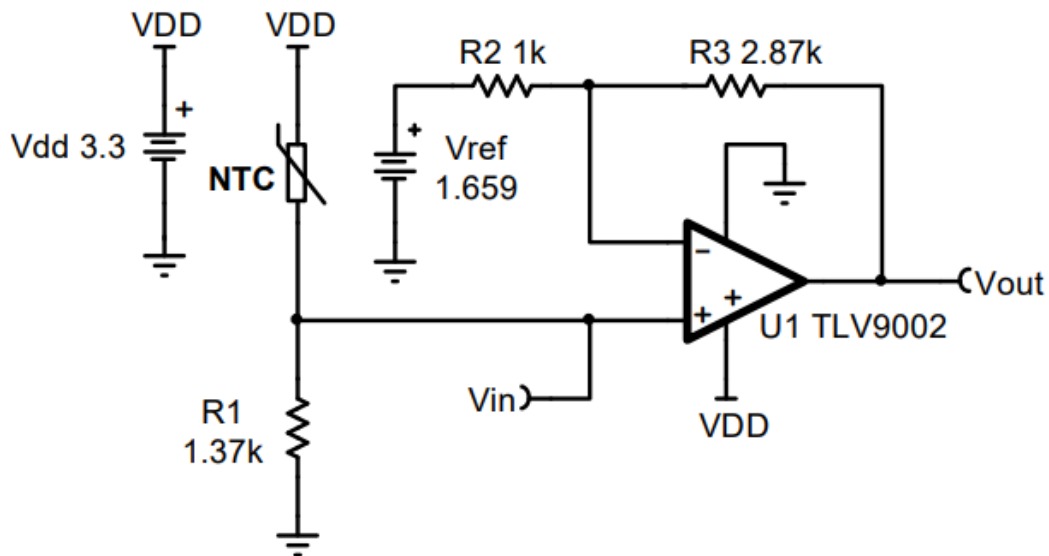
General Purpose Amplifiers

www.ti.com/general-amps

www.ti.com/circuitcookbooks



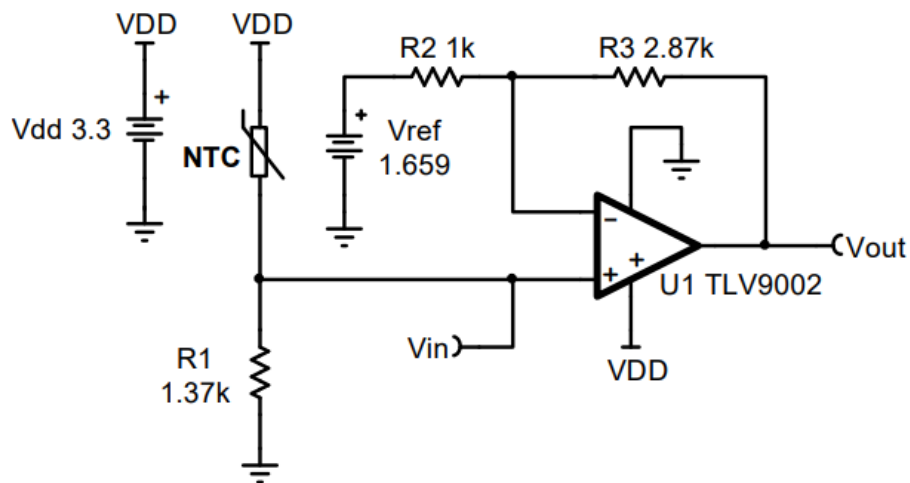
Circuit Description



$$V_O = VDD \times \frac{R1}{R1 + R_{NTC}} \times \frac{R3 + R2}{R2} - V_{ref} \times \frac{R3}{R2}$$

Design Goals

Temperature		Output		Supply		
T_{Min}	T_{max}	V_{outMin}	V_{outMax}	V_{dd}	V_{ee}	V_{ref}
25 °C	50 °C	50mV	3.25V	3.3V	0V	1.659V



NTC Voltage
Divider, V_{in}

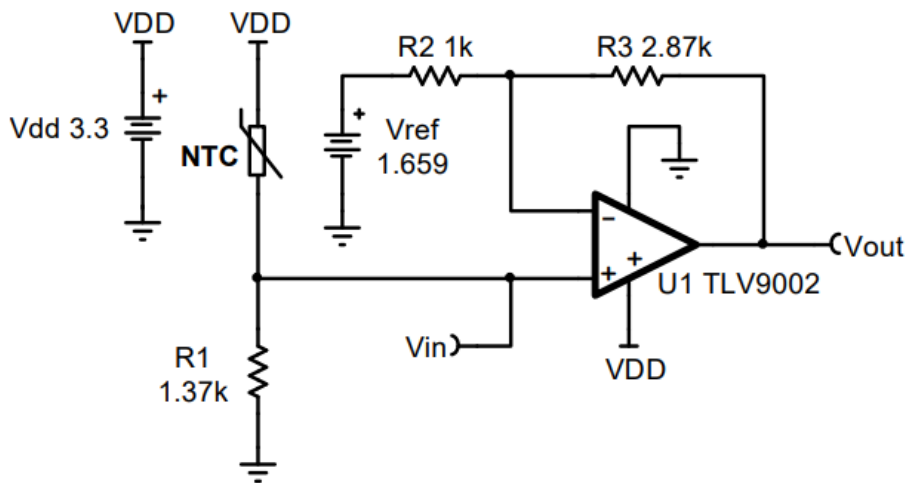
Inverting gain
times V_{ref}

$$V_O = VDD \times \frac{R1}{R1 + R_{NTC}} \times \frac{R3 + R2}{R2} + V_{\text{ref}} \times \left(-\frac{R3}{R2} \right)$$

Non-inverting
amplifier gain

Design Steps

Temperature		Output		Supply		
T_{Min}	T_{max}	V_{outMin}	V_{outMax}	V_{dd}	V_{ee}	V_{ref}
25 °C	50 °C	50mV	3.25V	3.3V	0V	1.659V



$$R_{NTC_max} = R_{NTC@25^{\circ}C} = 2.252 \text{ k}\Omega$$

$$R_{NTC_min} = R_{NTC@50^{\circ}C} = 819.7 \text{ }\Omega$$

$$R1 = \sqrt{R_{NTC@25^{\circ}C} + R_{NTC@50^{\circ}C}}$$

$$R1 = \sqrt{2.252 \text{ k}\Omega + 819.7 \text{ }\Omega} = 1.359 \text{ k}\Omega$$

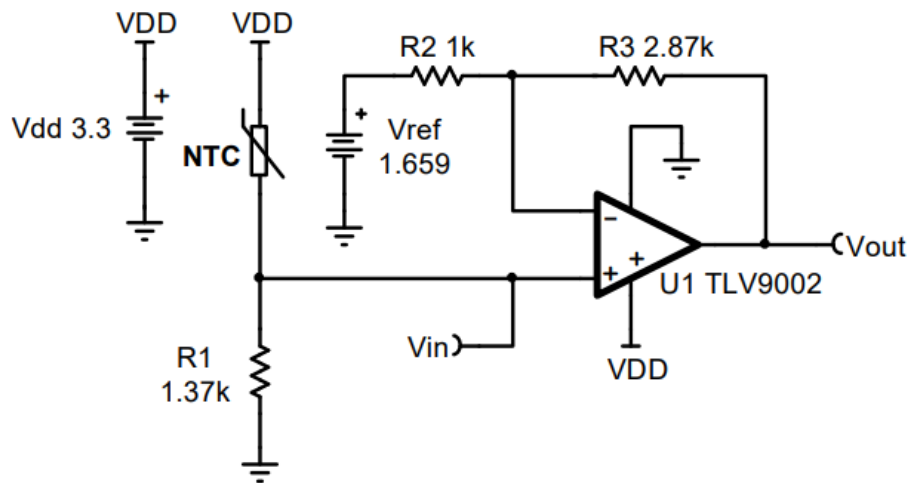
its multiply not add, its the geometric mean

$$V_{inMin} = V_{dd} \times \frac{R1}{R_{NTC_max} + R1} = 1.248V$$

$$V_{inMax} = V_{dd} \times \frac{R1}{R_{NTC_min} + R1} = 2.065V$$

Design Steps

Temperature		Output		Supply		
T_{Min}	T_{max}	V_{outMin}	V_{outMax}	V_{dd}	V_{ee}	V_{ref}
25 °C	50 °C	50mV	3.25V	3.3V	0V	1.659V



$$Gain_{ideal} = \frac{V_{outMax} - V_{outMin}}{V_{inMax} - V_{inMin}} = 3.917V/V$$

$$Gain = \frac{R2 + R3}{R2}$$

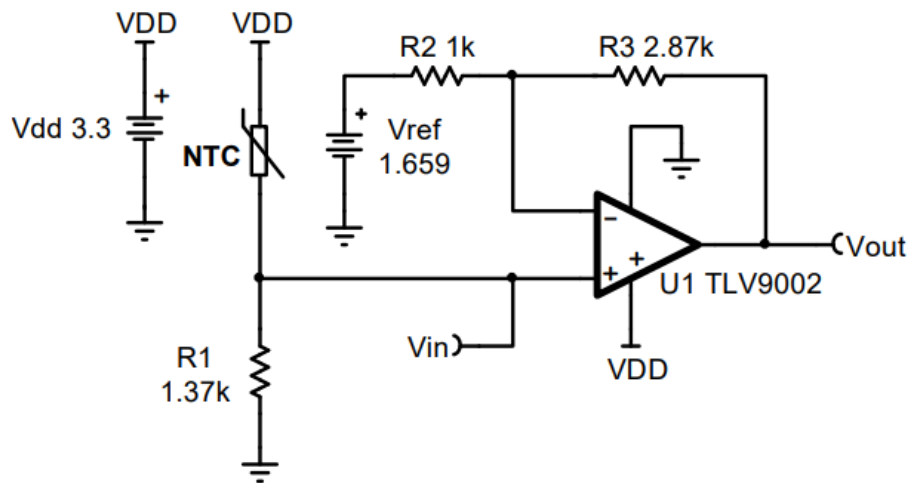
$$R2 = 1\text{ k}\Omega$$

$$R3 = R2 \times (Gain_{ideal} - 1) = 2.917\text{ k}\Omega$$

$$Gain_{actual} = \frac{R2 + R3}{R2} = 3.87V/V$$

Design Steps

Temperature		Output		Supply		
T_{Min}	T_{max}	V_{outMin}	V_{outMax}	V_{dd}	V_{ee}	V_{ref}
25 °C	50 °C	50mV	3.25V	3.3V	0V	1.659V



$$V_{o_swing} = (V_{inMax} - V_{inMin}) \times \text{Gain_actual}$$

$$V_{o_swing} = (2.065 \text{ V} - 1.248 \text{ V}) \times 3.87 \frac{\text{V}}{\text{V}} = 3.162 \text{ V}$$

$$V_{oMax} = V_{mid - supply} + \frac{V_{o_swing}}{2}$$

$$V_{oMax} = 1.65 \text{ V} + \frac{3.162 \text{ V}}{2} = 3.231 \text{ V}$$

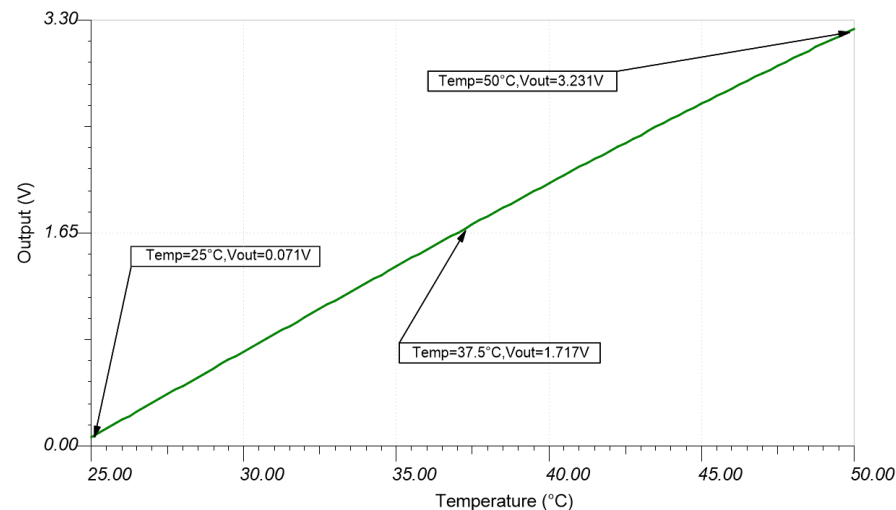
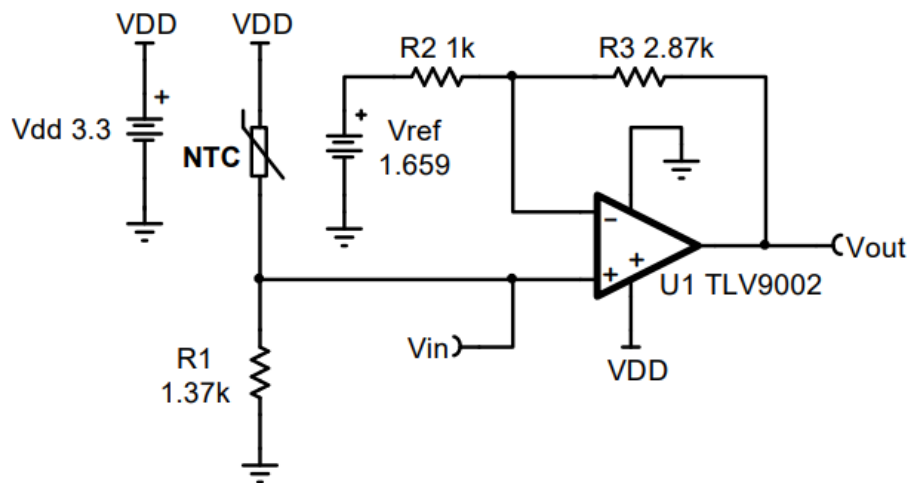
$$V_{oMax} = V_{inMax} \times \text{Gain_actual} - V_{ref} \times \frac{R3}{R2}$$

$$V_{ref} = \frac{2.065 \text{ V} \times 3.87 \frac{\text{V}}{\text{V}} - 3.231 \text{ V}}{\frac{2.87 \text{ k}\Omega}{1 \text{ k}\Omega}} = 1.659 \text{ V}$$

this is to choose vref
such that the swing is symmetric
about half supply rail

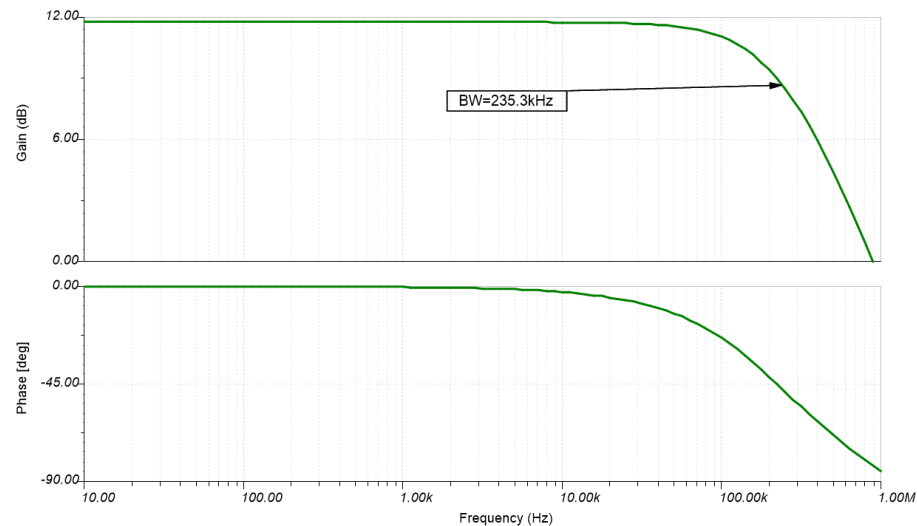
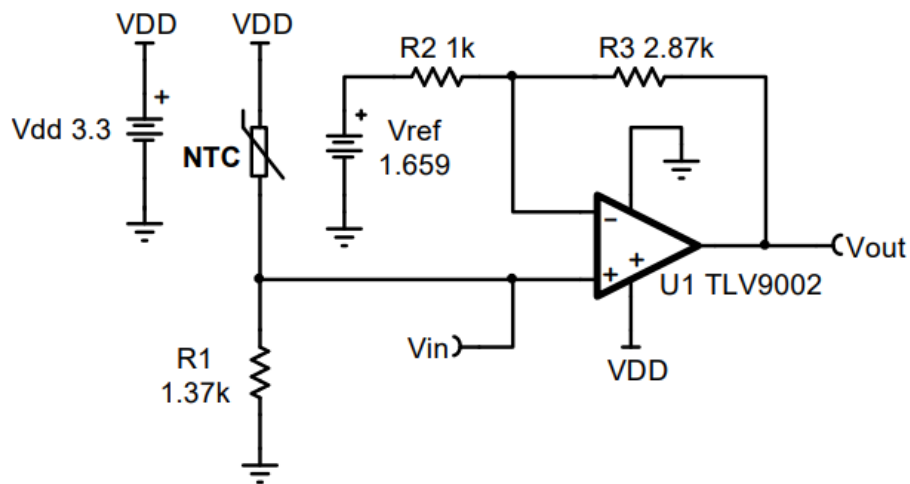
DC Results

Temperature		Output		Supply		
T_{Min}	T_{max}	V_{outMin}	V_{outMax}	V_{dd}	V_{ee}	V_{ref}
25 °C	50 °C	50mV	3.25V	3.3V	0V	1.659V



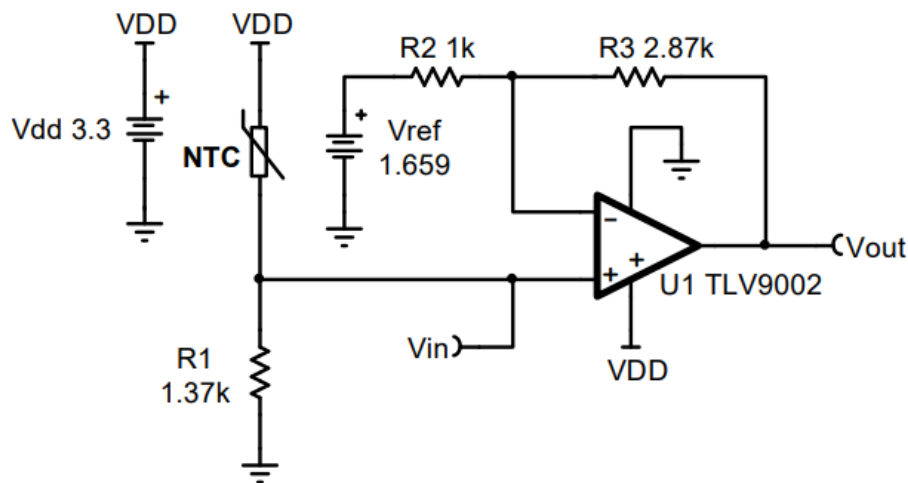
AC Results

Temperature		Output		Supply		
T_{Min}	T_{max}	V_{outMin}	V_{outMax}	V_{dd}	V_{ee}	V_{ref}
25 °C	50 °C	50mV	3.25V	3.3V	0V	1.659V



Design Notes

Temperature		Output		Supply		
T_{Min}	T_{max}	V_{outMin}	V_{outMax}	V_{dd}	V_{ee}	V_{ref}
25 °C	50 °C	50mV	3.25V	3.3V	0V	1.659V



Design Notes:

1. For temperature sensing using an NTC thermistor, the resistor, R1, is chosen based on the temperature range and the NTC's value.
2. Operate within the linear output voltage swing (See A_{OL} specification) to minimize non-linearity errors.
3. The reference voltage, Vref, can be created using a DAC or voltage divider. If a voltage divider is used the equivalent resistance of the voltage divider will influence the gain of the circuit.

Design Resources

EE Cookbook: Op Amp

www.ti.com/circuitcookbooks

Step-by-step circuit design of common op amp building block circuits.

TI Designs

www.TI.com/tidesigns

Ready-to-use reference designs with theory, calculations, simulations schematics, PCB files, bench test results

Analog Engineer's Pocket Reference

www.TI.com/analogrefguide

PDF, iTunes app and hardcopy available
PCB, analog, mixed signal design formulae
Conversions, tables, equations

TI Precision Labs

www.TI.com/precisionlabs

Quiz questions, problems, solutions
Labs and evaluation module (EVM) available

TINA-TI™ simulation software

www.TI.com/tool/tina-ti

Complete SPICE simulator DC, AC, transient, noise analysis
Schematic entry and post-processor for waveform math

DIYAMP-EVM

www.TI.com/DIYAMP-EVM

Evaluation module providing engineers with SC70, SOT23, SOIC packaging and 12 popular amplifier configurations

The Signal

www.TI.com/signalbook

PDF, iTunes app and hardcopy available
A compendium of blog posts on op amp design topics including offset voltage, input bias current, stability, noise and more

Analog Wire Blog

www.TI.com/analogwire

Technical blogs written by analog experts
Tips, tricks, and design techniques

TI E2E™ Community

www.TI.com/e2e

Support forums for all TI products

Op Amp Parametric Quick Search

www.TI.com/amplifiers

Search for precision, high-speed, general-purpose, ultra-low-power, audio and power op amps

Op Amp Parametric Cross-Reference

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Find similar TI op amps using competitive part numbers

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